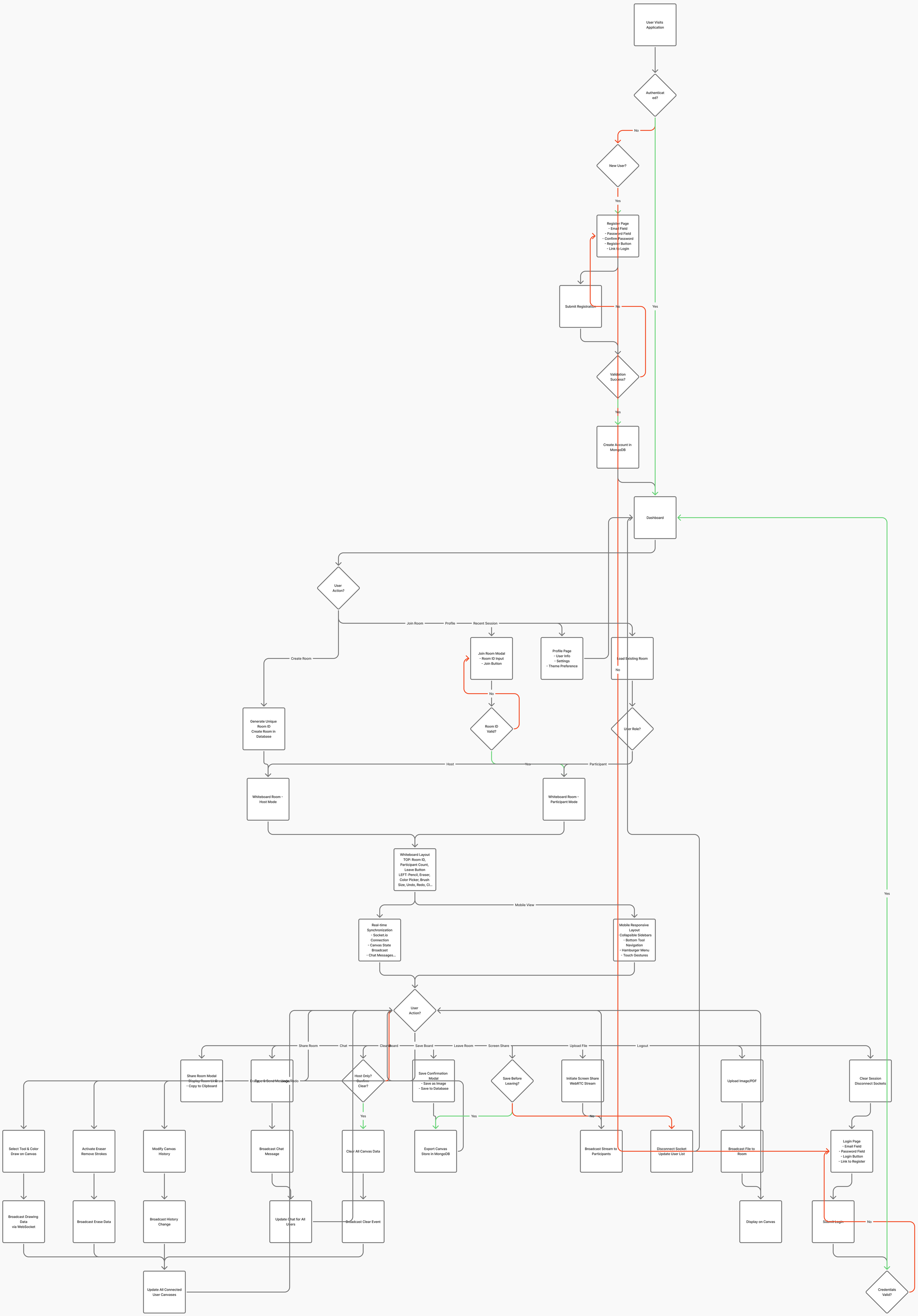
Flow chart


User Visits Application
Authenticated?
No
New User?
Yes
Register Page
- Email Field
- Password Field
- Confirm Password
- Register Button
- Link to Login
Submit Registration
No
Yes
Validation Success?
Yes
Create Account in MongoDB
Dashboard
User Action?
Join Room
Profile
Recent Session
Create Room
Join Room Modal
- Room ID Input
- Join Button
Profile Page
- User Info
- Settings
- Theme Preference
Load Existing Room
No
Generate Unique Room ID
Create Room in Database
No
Room ID Valid?
User Role?
Host
Yes
Participant
Whiteboard Room - Host Mode
Whiteboard Room - Participant Mode
Whiteboard Layout
TOP: Room ID, Participant Count, Leave Button
LEFT: Pencil, Eraser, Color Picker, Brush Size, Undo, Redo, Cl...
Mobile View
Real-time Synchronization
- Socket.io Connection
- Canvas State Broadcast
- Chat Messages...
Mobile Responsive Layout
Collapsible Sidebars
- Bottom Tool Navigation
- Hamburger Menu
- Touch Gestures
Yes
User Action?
Share Room
Chat
Clear Board
Save Board
Leave Room
Screen Share
Upload File
Logout
Share Room Modal
Display Room Link
- Copy to Clipboard
Draw
Erase
Type & Send Message
Undo/Redo
Host Only? Confirm Clear?
Save Confirmation Modal
- Save as Image
- Save to Database
Save Before Leaving?
Initiate Screen Share WebRTC Stream
Upload Image/PDF
Clear Session Disconnect Sockets
Yes
Yes
No
Select Tool & Color Draw on Canvas
Activate Eraser Remove Strokes
Modify Canvas History
Broadcast Chat Message
Clear All Canvas Data
Export Canvas Store in MongoDB
Broadcast Stream to Participants
Disconnect Socket Update User List
Broadcast File to Room
Login Page
- Email Field
- Password Field
- Login Button
- Link to Register
Broadcast Drawing Data via WebSocket
Broadcast Erase Data
Broadcast History Change
Update Chat for All Users
Broadcast Clear Event
Display on Canvas
Submit Login
No
Update All Connected User Canvases
Credentials Valid?